

Does Quantization Erode Grounding?

A Faithfulness Audit of On-Device LLM Compression

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Abstract

Weight quantization is the default route to running language models on consumer hardware, yet its effect on *contextual faithfulness*—a model’s willingness to ground answers in provided context rather than memorized priors—remains untested: faithfulness benchmarks use full-precision models, and quantization studies measure accuracy and perplexity, not grounding. We audit two instruction-tuned 3B models (Qwen2.5-3B, Llama-3.2-3B) across seven GGUF precisions from 16-bit to 2-bit, under aligned, counterfactual, and unanswerable contexts, using 300 SQuAD-derived items and three prompt templates per condition (900 trials per condition per precision). With an item-level clustered bootstrap we detect no difference from 16-bit at 8, 6, or 5 bits in either model; below that, declines appear whose onset is *model-dependent* (4-bit for Llama, 3-bit for Qwen). Rates of producing the original, context-contradicting answer stay low throughout, with no consistent increase as precision falls, so we find no support for the intuitive “quantization erodes grounding” mechanism: the faithfulness that is lost appears as off-target responses instead. Qwen additionally collapses into incoherent output at 2-bit while Llama stays fluent, making extreme compression a per-model question. Code and harness: <https://github.com/movvaneeraj/quantfaith>.

1 Introduction

Running large language models locally—on laptops, phones, and edge devices—almost always means quantizing them: storing weights at 8, 4, or even 2 bits instead of 16. Community toolchains (llama.cpp, GGUF k-quants, Ollama) have made this a one-command operation, and a large literature characterizes what it costs in perplexity and task accuracy (Frantar et al., 2023; Lin et al., 2024; Dettmers et al., 2022; Lee et al., 2024; Kurt, 2026). The consensus is encouraging on average

but nuanced at the margins: quantized models often match full precision on multiple-choice knowledge benchmarks, yet degrade disproportionately on subtler behaviors such as instruction-following and hallucination detection (Kurt, 2026; Lee et al., 2024).

Contextual faithfulness—whether a model answers from the context it is given rather than from memorized parametric knowledge—is exactly the kind of subtle behavior that aggregate accuracy can hide, and it is safety-critical for the on-device retrieval-augmented and tool-using systems that quantization is meant to enable. A model that quietly ignores a retrieved document and “corrects” it with a stale memorized fact is unfaithful even when it is fluent and confident (Longpre et al., 2021; Zhou et al., 2023; Xie et al., 2024; Ming et al., 2025). The faithfulness literature studies this almost exclusively at full precision, and the quantization literature never measures grounding under knowledge conflict. The question at the intersection has gone unasked: **does compressing a model change how it arbitrates between context and memory?** There is a mechanistic reason to expect that it might. If low-bit weight perturbation damaged a context-reading pathway more than a memory-retrieval pathway, compressed models would fall back on memorized priors more often—an *erosion of grounding* invisible to standard accuracy.

We test this directly. Treating weight precision as the independent variable, we sweep two 3B instruction-tuned models across seven GGUF precisions (16, 8, 6, 5, 4, 3, 2 bits) in three context regimes—**aligned**, **counterfactual**, and **unanswerable** (§3)—and measure how each arbitrates between context and memory. Our contributions:

1. The first audit of contextual faithfulness with *bit-width as the independent variable*, across two model families and three context regimes,

with item-level bootstrap intervals.

2. *A negative result on the natural hypothesis:* we detect no change in faithfulness down to 5-bit, and original-answer rates stay low with no consistent increase as precision falls—we find no evidence that compression pushes models back toward their priors.
3. *A characterization of the low-bit cliff* as model-dependent: declines begin at different precisions in the two models, and at 2-bit one collapses into incoherence while the other stays fluent, so “how low can you go” cannot be read off a single checkpoint.

2 Related Work

Contextual faithfulness and knowledge conflict.

A model is *contextually faithful* when it grounds answers in supplied context rather than parametric memory. Longpre et al. (2021) introduced the entity-substitution paradigm for inducing knowledge conflict—replacing the answer entity in a passage with a plausible alternative—which we adopt. Xie et al. (2024) show that under conflict LLMs exhibit a strong confirmation bias toward parametric memory, and Ming et al. (2025) demonstrate with FaithEval that even strong models struggle with counterfactual contexts. Proposed remedies include opinion-based and counterfactual prompting (Zhou et al., 2023), counterfactual fine-tuning (Neeman et al., 2023), and preference alignment over context-faithful responses (Bi et al., 2025). All operate at full precision.

Quantization of LLMs. Post-training weight quantization (Frantar et al., 2023; Lin et al., 2024; Dettmers et al., 2022) and the GGUF k-quant family used by `llama.cpp` are now standard for local deployment. Recent unified evaluations report that quantized models track full precision on knowledge benchmarks but suffer disproportionately on instruction-following and hallucination detection, and that 2-bit formats can break down entirely (Kurt, 2026; Lee et al., 2024); work on code generation documents 2-bit models lapsing into “persistent hallucinations” and incoherent, repetitive output (Nyamsuren, 2024)—the same degeneration we quantify for Qwen2.5-3B. These studies measure accuracy, perplexity, and throughput; none measures contextual faithfulness under knowledge conflict.

3 Method

Conditions. Each item places a question against a context in one of three regimes. **Aligned:** the context states the true answer; a control measuring whether the model can read the answer at all, scored as *accuracy*. **Counterfactual:** following Longpre et al. (2021), the gold answer entity in the passage is replaced by a plausible but false substitute, so a *faithful* model returns the substituted answer. **Unanswerable:** the context does not contain the answer, so a faithful model abstains rather than fabricating.

Data. We build all items from SQuAD v2.0 (Rajpurkar et al., 2016, 2018), which supplies both answerable passage-question-answer triples and human-curated unanswerable questions. For counterfactual items we replace the gold answer span with a different entity-like answer drawn at random from the pool of answers elsewhere in the corpus, excluding strings already present in the passage. Unlike Longpre et al. (2021), our sampler does *not* enforce type consistency: a location answer may be replaced by a person (§6). We use 300 items per condition and three paraphrased prompt templates per item, so that a measured trend reflects model behavior rather than one phrasing; this yields 900 *trials* (not independent items) per condition per precision.

Models and precisions. We evaluate two instruction-tuned 3B models from different families, **Qwen2.5-3B-Instruct** (Qwen Team, 2024) and **Llama-3.2-3B-Instruct** (Llama Team, AI @ Meta, 2024), each served through Ollama at seven GGUF precisions: `fp16` (16-bit), `q8_0`, `q6_K`, `q5_K_M`, `q4_K_M`, `q3_K_M`, and `q2_K`. Decoding is greedy (temperature 0) with a fixed seed and a 128-token cap, served by Ollama 0.12 on a single NVIDIA T4; all factors other than precision are held constant.

Scoring. Each response is mapped to a label by normalized, token-contiguous matching against the context answer, the original SQuAD answer, and a curated list of abstention cues. For counterfactual items the labels are `CONTEXT` (faithful), `ORIGINAL`, `ABSTAIN`, `AMBIGUOUS`, and `OTHER`. We deliberately call the second label `ORIGINAL` rather than *parametric*: we did not run a closed-book probe, so emitting the original answer shows that the model resisted the context, not that the fact was retrievable from memory (§6). When a response

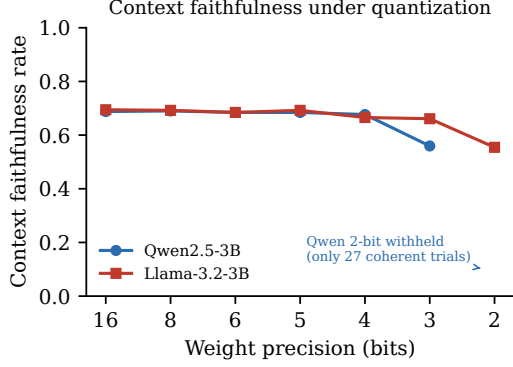


Figure 1: Context faithfulness rate (over coherent trials) versus weight precision. The two models track each other from 16 down to 5 bits; below that they separate, with Llama declining from 4-bit and Qwen from 3-bit (Table 1). Qwen’s 2-bit point is withheld: only 27 of 900 trials remain coherent, too few for a stable rate.

surfaces *both* candidate answers (e.g. “the passage says X, though really Y”), we assign AMBIGUOUS, crediting neither: correction framing makes any first-mention heuristic unreliable. Heavily quantized models can also emit non-linguistic, degenerate output. We flag a response as DEGENERATE if it is empty, contains a run of ≥ 8 identical characters, or is at least 15 characters long with $< 40\%$ alphabetic content; this character-level heuristic does not catch word-level repetition (§6). Counterfactual and unanswerable rates are computed over the *coherent* trials only, with the coherence rate reported separately, so that a model’s collapse is not silently scored as a grounding decision; aligned accuracy is computed over all trials, so a degenerate output counts as incorrect.

4 Results

No detectable change to 5-bit; declines below. We estimate uncertainty with an item-level clustered bootstrap (2000 resamples) that resamples the 300 items, keeping an item’s three templates together, since templates are repeated measurements rather than independent observations. Fig. 1 plots the headline result and Table 1 reports each precision’s rate and its *paired* difference from 16-bit. For both models the 8-, 6-, and 5-bit differences are small and their intervals include zero, so we detect no difference from 16-bit at those precisions; with 300 items the intervals are too wide to establish equivalence, and we do not claim it. Below 5 bits the picture is model-dependent: Llama-3.2-3B shows a small but detectable decline already at

Bits	Faith.	Δ vs. 16-bit [95% CI]	Orig.	Coh.	Acc.
<i>Qwen2.5-3B-Instruct</i>					
16	0.688	—	0.053	1.00	0.923
8	0.690	+0.002 [−0.008, +0.012]	0.052	1.00	0.927
6	0.684	−0.003 [−0.018, +0.011]	0.053	1.00	0.921
5	0.684	−0.003 [−0.023, +0.017]	0.054	1.00	0.927
4	0.677	−0.011 [−0.037, +0.016]	0.048	1.00	0.914
3	0.560	−0.128 [−0.172, −0.086]	0.059	1.00	0.868
2	—	withheld ($n=17$ items)	—	0.03	0.000
<i>Llama-3.2-3B-Instruct</i>					
16	0.694	—	0.057	1.00	0.943
8	0.692	−0.002 [−0.011, +0.006]	0.058	1.00	0.942
6	0.684	−0.010 [−0.022, +0.001]	0.057	1.00	0.938
5	0.692	−0.002 [−0.018, +0.013]	0.059	1.00	0.944
4	0.666	−0.029 [−0.052, −0.008]	0.058	1.00	0.938
3	0.661	−0.033 [−0.060, −0.007]	0.063	1.00	0.916
2	0.554	−0.140 [−0.180, −0.103]	0.057	1.00	0.891

Table 1: Per-precision summary. **Faith.** = context-faithfulness and **Orig.** = original-answer rate, both over coherent counterfactual trials; **Coh.** = fraction of counterfactual trials that are non-degenerate; **Acc.** = aligned accuracy over all trials. Δ is the paired difference in faithfulness from 16-bit with a 95% item-level clustered bootstrap interval (2000 resamples over the 300 items, an item’s three templates resampled together), computed on items coherent at both precisions; **bold** marks intervals excluding zero. Qwen’s 2-bit behavioral rates are withheld: only 27 of 900 trials (17 items) are coherent.

4-bit (−0.029, CI [−0.052, −0.008]), whereas for Qwen2.5-3B we detect no difference at 4-bit and a sharp drop at 3-bit (−0.128, CI [−0.172, −0.086]). The common claim that 4-bit is “free” therefore does not hold uniformly: it is mildly costly for one of our two models.

Original-answer rates show no consistent rise.

If compression pushed models toward memorized priors, the ORIGINAL rate would rise as precision drops. It does not rise consistently: it stays between 0.048 and 0.063 in both models at every precision with a coherent output distribution, and every paired difference from 16-bit has an interval containing zero (Table 1; Fig. 2b). Faithfulness lost at low precision does not reappear as the contradicted original answer; it appears instead as off-target (OTHER) responses, which rise from $\approx 20\%$ at 16-bit to 28.6% (Qwen, 3-bit) and 35.0% (Llama, 2-bit). This is the clearest negative result in our study.

The 2-bit cliff is model-dependent. At 2-bit the two models diverge completely. Qwen2.5-3B collapses: 873/900 counterfactual outputs are degenerate, leaving only 27 coherent trials across 17 distinct items, too few for a stable rate, so we

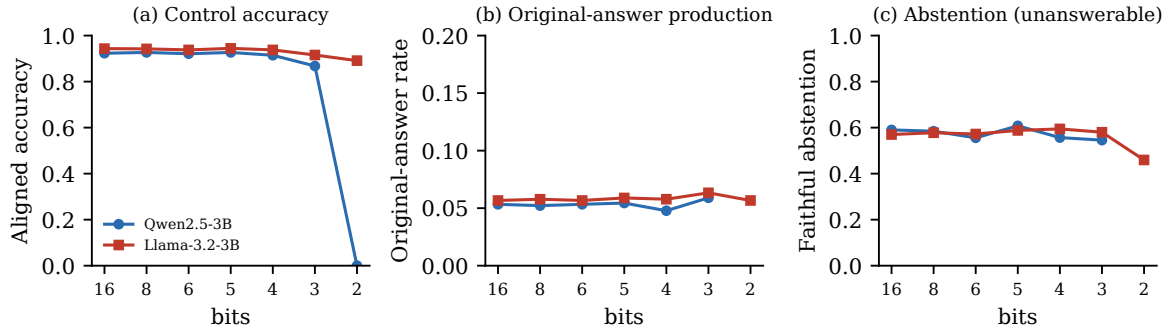


Figure 2: (a) Aligned accuracy, (b) rate of producing the original (context-contradicting) answer, and (c) faithful-abstention rate versus precision. Panel (b) stays between 0.048 and 0.063 at every precision in both models. Abstention (c) fluctuates in both directions with no monotone trend. Panels (b) and (c) omit points with fewer than 30 coherent trials (Qwen at 2-bit); panel (a) is computed over all trials, so Qwen’s 2-bit value of 0.00 includes its degenerate outputs.

withhold its 2-bit behavioral estimates; its aligned accuracy falls to 0.00. Llama-3.2-3B never collapses: it stays fully coherent at 2-bit and still reaches 0.554 faithfulness (-0.140 from 16-bit, CI $[-0.180, -0.103]$) with 0.891 aligned accuracy. The same nominal `q2_K` format is catastrophic for one 3B model and merely lossy for the other.

Coherence does not explain the general decline.

The faithfulness cliff is not a coherence cliff. Llama loses 14.0 points of faithfulness between 16- and 2-bit while its coherence rate stays at 1.00, and Qwen loses 12.8 points at 3-bit while still fully coherent: loss of coherence explains exactly one cell of our design, Qwen at 2-bit. Aligned accuracy (Fig. 2a) moves in the same direction but by less—from 16- to 2-bit Llama loses 0.052 accuracy against 0.140 faithfulness, and Qwen at 3-bit loses 0.056 against 0.128. The two are measured on different tasks with different ceilings, so we report the gap without inferring how much of the faithfulness loss general capability degradation explains.

Abstention is noisy rather than flat. On unanswerable contexts both models abstain only about 46–61% of the time and fabricate an answer otherwise, a weak baseline consistent with prior reports for small models. Against 16-bit, several precisions differ detectably in both directions (Qwen lower at 6-, 4-, and 3-bit; Llama higher at 5- and 4-bit and markedly lower at 2-bit, -0.110 , CI $[-0.153, -0.068]$), with no monotone trend. Abstention is therefore unstable under quantization rather than flat.

5 Discussion and Conclusion

We find no sign that quantization erodes grounding in the mechanistically intuitive way: in neither model does the original-answer rate rise consistently as precision falls. This is the finding we hold with most confidence. What does change is harder to summarize. We detect no change in faithfulness down to 5-bit, but the onset of decline below that is model-specific—4-bit in Llama-3.2-3B, 3-bit in Qwen2.5-3B—and the 2-bit divergence is starker still: the same `q2_K` format leaves Llama-3.2-3B fluent and mildly degraded while Qwen2.5-3B produces almost no usable language at all. Practitioners should not read a single safe precision off one checkpoint.

We resist two tempting simplifications. The decline is not merely loss of fluency: outside Qwen at 2-bit, faithfulness falls while coherence remains perfect. It also falls two to three times as far as aligned accuracy over the same range, though the two are not measured on the same task and we cannot say how much of that gap general capability loss explains. The conflict setting appears more fragile than plain reading comprehension; why Llama-3.2-3B tolerates 2-bit where Qwen2.5-3B does not—architecture, tokenizer, training data, or quantization sensitivity—remains open. For on-device deployment: we detected no faithfulness cost at 5-bit and above, though our intervals cannot rule out a small one; 4-bit and below should be validated per model, not assumed safe; and faithfulness should be audited alongside accuracy, not presumed to track it.

6 Limitations

No closed-book baseline. We never query the models without context, so we cannot show that the original SQuAD answer is knowledge each model could actually retrieve. Our ORIGINAL rate describes how often the original answer is produced, not how often accessible parametric memory is used, and string matching can miss paraphrases of that answer, so we do not treat it as a bound in either direction. The central negative result should be read accordingly: we observe no increase in the behavior that memory-reversion would produce. A closed-book pass that filters items to those a model answers correctly without context is the obvious next experiment, and we could not run it in the revision window.

Dataset construction. Our counterfactual substitutions are drawn at random from a single pool of entity-like answers and are *not* type-checked, so a location answer can be replaced by a person (for example, a passage asserting that Normandy is in “Saddam Hussein”). Implausible substitutions may make the counterfactual easier to reject and thus depress measured faithfulness. The procedure is held constant across precisions, but we cannot rule out that it interacts with precision—a model may reject an implausible substitution at different rates as its capability degrades—so the effect on our trends, and not only on our levels, is unquantified. It also limits comparison of our absolute rates against benchmarks built with type-constrained substitution.

Scoring. Labels come from string matching, which we audited and found to have three failure modes; Appendix A quantifies each. (i) A target string is credited wherever it appears, including under negation, so “Paris is not the answer” scores as answering Paris. (ii) An abstention cue is checked before answer content, so “the context does not say, but it was founded in 1998” is scored as abstention. (iii) The coherence detector is character-level and misses word-level repetition loops, which we find to be rare enough not to affect the reported coherence rates. The first two mechanisms can move labels in both directions—cues and paraphrases outside our lists are missed as well as over-applied—so all reported rates, including aligned accuracy, should be read as heuristic estimates subject to scoring error rather than as bounds. This audit is automatic; we did not manually annotate a strati-

fied sample, and an LLM-judge or human rescoring pass would tighten all three.

Scope. We evaluate one checkpoint from each of two families, so we cannot separate architecture from tokenizer, training data, and quantization sensitivity: the divergence we report is model-dependent, and motivates rather than establishes a general taxonomy. Larger models may behave differently. We use a single quantization toolchain (GGUF k-quants via `llama.cpp`/Ollama); GPTQ, AWQ, and activation quantization may differ. A substantial share of counterfactual responses fall in the OTHER category (Appendix A), reflecting both genuine off-target answers and imperfect automatic construction. We hold decoding greedy; sampling temperatures could interact with low-bit degeneration. Finally, our intervals are per-comparison and we make no multiplicity correction, so borderline results such as Llama at 4-bit should be read as suggestive rather than definitive.

Use of AI Assistants

We used an AI assistant (Anthropic’s Claude) in the course of this work, including in writing and revising the manuscript and in preparing, checking, and presenting the code and analysis. The research questions, experimental design, and all claims and their interpretation are the authors’ own, and the authors take full responsibility for the content of this paper, including any errors.

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A Scoring Audit

This appendix gives the numbers behind the scoring failure modes discussed in §6. All counts are over the full 18,900 trials per model. The audit is automatic, based on keyword presence, and is itself approximate; it is intended to bound the order of magnitude of each failure mode, not to replace manual annotation.

Credit under negation. 4.6–6.1% of CORRECT labels and 11.5–11.9% of CONTEXT labels (ranges across the two models) contain a negation token somewhere in the response and may therefore be over-credited.

Abstention cues before answer content. 15.4% (Qwen) and 30.5% (Llama) of ABSTAIN labels on unanswerable items contain a contrastive connector such as “but” or “however”, indicating a response that hedges and then answers anyway.

Word-level repetition. The character-level degeneracy detector misses word-level repetition loops. Manual inspection finds 5 such cases among 16,271 non-degenerate Qwen outputs and 0 among 18,899 for Llama, so the reported coherence rates are essentially unaffected.

Label mix on counterfactual items. Over coherent counterfactual trials pooled across precisions, OTHER accounts for 22.6% (Qwen) and 22.0% (Llama) of responses, ABSTAIN for 3.3% and 3.1%, and AMBIGUOUS for 2.7% in both models.